S.No. : 59

NEE 4101

No. of Printed Pages : 05

Following Paper ID and Roll No. to be filled in your Answer Book.											
PAPER ID : 43301	Roll No.	1	2	4	0	4	3	3	3	9	2

B. Tech. Examination, 2024-25

(Odd Semester)

BASIC ELECTRICAL ENGINEERING

Time : Three Hours] *[Maximum Marks : 60*

Note :– Attempt all questions.

SECTION–A

1. Attempt all parts of the following : $8\times1=8$

(a) What is the difference between mesh and loop?

(b) What is the difference between linear and non-linear elements?

(c) Define impedances.

(d) Define line voltage.

(e) What is Lenz's law?

[P. T. O.

(f) Define eddy current loss.

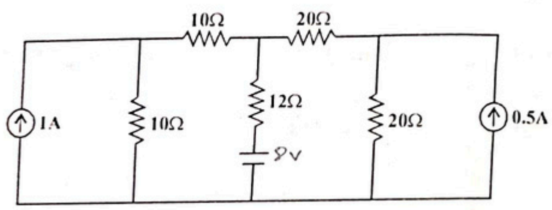
(g) Define slip.

(h) What is the synchronous speed?

SECTION–B

2. Attempt any two parts of the following : $2\times6=12$

(a) Find the current in 12 Ω resistance using Thevenin's theorem :



(b) Explain the measurement of three phase power by two wattmeter method.

(c) Draw and explain the working of induction type energy meter.

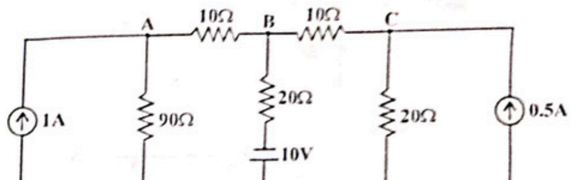
(d) Why the synchronous meter is not self starting?

SECTION–C

Note :– Attempt all questions. Attempt any two parts from each questions. $8\times5=40$

3. (a) State and explain maximum power transfer theorem to solve network problems and also write two applications.

(b) Find the current in each element by nodal analysis :



(c) How is delta connected network converted into equivalent star connected network?

4. (a) An a.c. ckt has four different voltages :

$V_1 = 100 \sin 500\, t$

$V_2 = 180 \sin (500\, t + \pi/3)$

$V_3 = -60 \cos 500\, t$

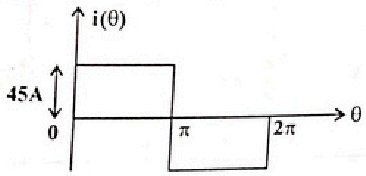
$V_4 = 150 \sin (500\, t + \pi/4)$

what will the equivalent voltage in that circuit also calculates r.m.s. average value.

(b) For LCR series circuit drive an expression for resonant frequnecy :

$$f_r = \sqrt{f_1 f_2}$$

(c) Calculate the r.m.s. and average value of the given waveform :



5. (a) Compare moving iron instrument with dynamometer instrument and which is the best one.

(b) Derive e.m.f. equation of transformer.

(c) Compare electrical circuit with magnetic circuit.

6. (a) A 4 pole, 1200 rpm generator with lap wound armature has 65 slots and 12 conductors per slot. The flux per pole is 0.02 wb. Calculate the e.m.f. generated.

(b) Why the single phase induction motor is not self starting write two methods to start the single phase induction motor?

(c) Derive the torque equation for D.C. motor.

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